

Recent Achievements in Project Implementation at Statistics Poland

Web data in statistical production system

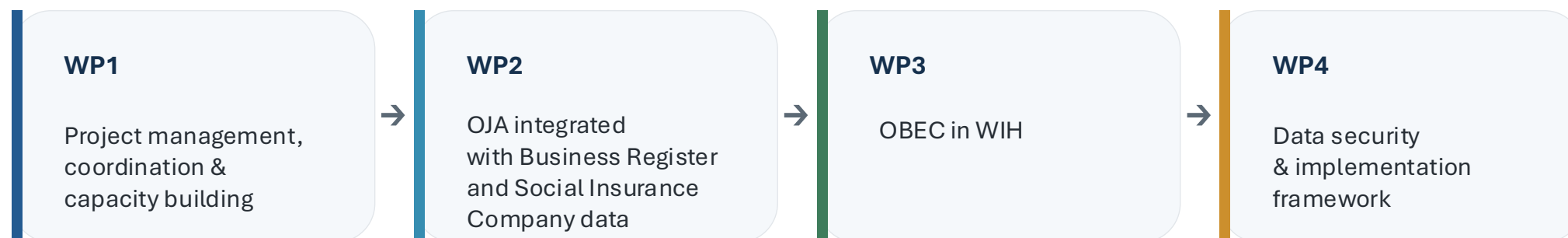
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Jacek Maślankowski
j.maslankowski@stat.gov.pl
Statistics Poland

Agenda

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|---|---------------------|-----|
| 1 | Project governance | WP1 |
| 2 | OJA data | WP2 |
| 3 | OBEC data | WP3 |
| 4 | Secure architecture | WP4 |

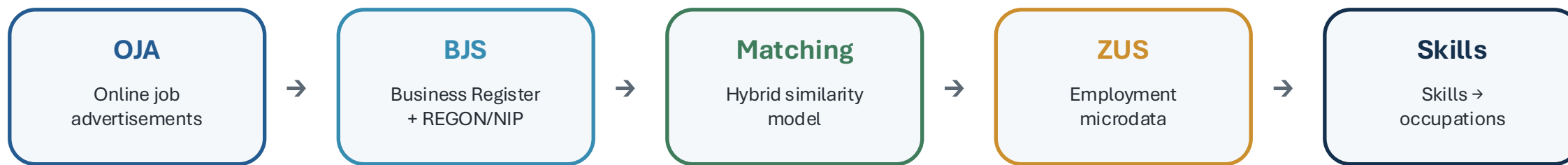
Clear message from WP1: WP2, 3 and 4 must use the WIH environment



WP2: general outline

- Goal: find skills of newly created jobs.
- Newly created jobs (not all jobs) identified through Social Insurance Company database, secured in Operational Micro Database.
- Eurostat OJA dataset is linked with Business Register through company names and additional variables.
- Eurostat OJA dataset is used for exploratory analysis, not as a statistical dataset.

WP2: from job advertisements to skills intelligence



Key methodological achievement

- Two-stage company-name matching: retrieve the 20 most similar candidates, then select the best candidate using token set ratio, Levenshtein distance and Jaro-Winkler similarity.
- The 50-25-25 hybrid weighting was selected for greater stability in ambiguous cases.
- The approach is designed for repeated use in subsequent years.

WP2: validation and scale have already produced substantial test outputs

27,820

Unambiguous matches

Companies identified through
validation

~8,000

Manual verification

Remaining ambiguous cases
(approx.)

620,174

July 2024 records

Records with an assigned REGON

72,606

Unique OJA / companies

72,606 ads published by 20,071 companies

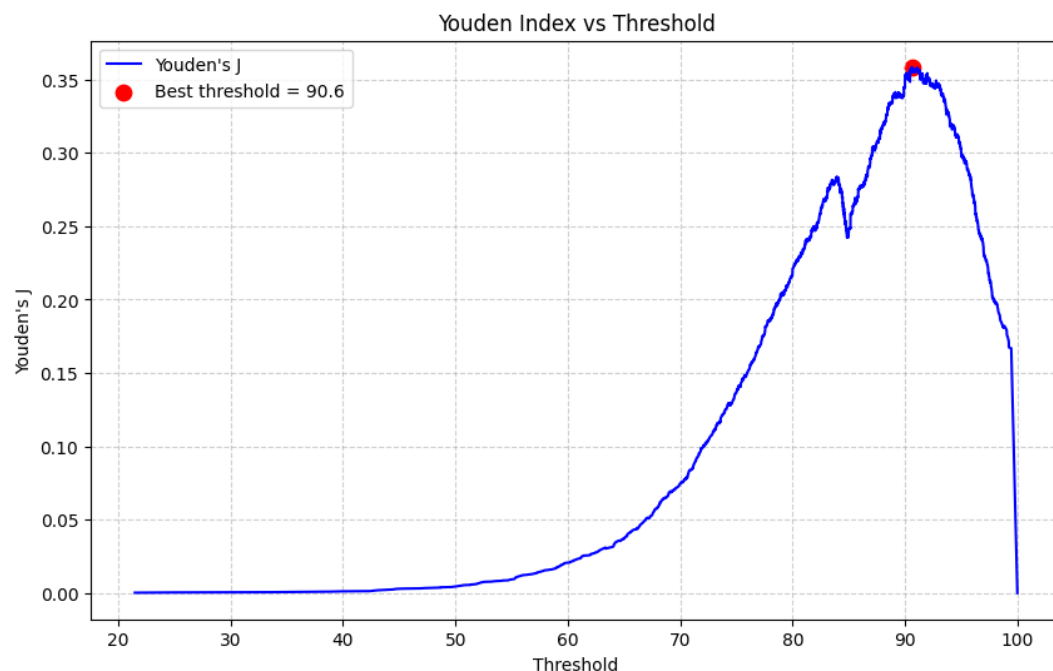
What the current stage demonstrates

- The REGON (Business ID) assignment methodology can be applied across the full set of 2024 monthly data.
- Linkage with ZUS (Social Insurance Company) is being conducted inside the closed Operational Microdata Base (OBM), with only aggregated outputs released externally.

WP2: Methodology at a Glance – matching company names in two datasets

The Youden index (J) is a classical measure of the quality of binary classification, used to identify the optimal cutoff point in predictive models.

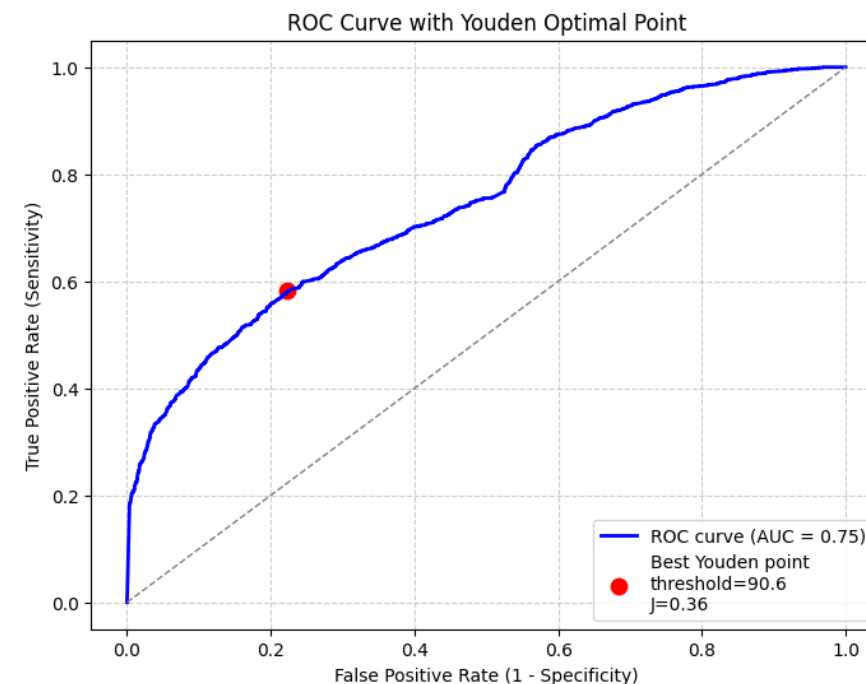
$$J = \text{sensitivity} + \text{specificity} - 1$$



Sensitivity - the probability of correctly identifying positives

Specificity - the probability of correctly rejecting negatives

Youden index was maximum at 90.6%



WP2: Methodology at a Glance – second step: linking OJA data with Social Insurance Company dataset

- in the Social Insurance Company dataset, within each company, employees were numbered in ascending order by the date of their first social insurance contribution (i.e., in order of hiring)
- in the Eurostat OJA dataset, within each company, job advertisements were sorted by *last_active_month* and numbered in an analogous way

The datasets were then linked by REGON (Business ID) and assigned sequential number.

Limitations

Advertisement no. 1 does not necessarily concern specifically the first, most recently hired employee.

Techniques such as fuzzy matching, Levenshtein distance and Jaro-Winkler similarity are sensitive to specific types of errors, like typos.

WP2: What is the value-added compared to the previous WIN project

The OJA dataset is not treated as a primary data source but as a supplementary dataset to another two administrative datasets (provided by Social Insurance Company and Statistics Poland)

The Eurostat OJA dataset is not treated as a statistical data source but exploratory dataset

Results are more reliable because the starting point is the administrative high quality dataset from Social Insurance Company

Data is de-duplicated based on the company names, e.g. for July 2024:
620 174 records with an identified and assigned company REGON number, 72 606 unique job advertisements, posted by 20 071 companies.

WP3: building the pipeline for internet data in business statistics

Current emphasis: acquire, verify, index and prepare enterprise web data for statistical use.

1 • Source acquisition

Company records from external yellow pages;
website URLs filtered and servers queried.
Company URLs from questionnaire.

2 • Content capture

Website metadata and body content
collected: headings, text, links,
multimedia and contacts.

3 • Processing

Cleaning, NACE
classification with ML/LLM approaches
and KRS financial-statement automation.

Planned / emerging applications

- Integration and mapping of official business registers to company website URLs.
- Development and evaluation of an NACE code classifier based on Large Language Models.
- Identification of web information that can support experimental statistics.

WP3: general tasks

- Acquisition and verification of corporate website URLs.
- Web scraping of corporate websites using the Web Intelligence Hub (WIH) or home environment to retrieve company descriptions.
- Development and evaluation of NACE code classifier using a Large Language Model.
- Crawling corporate websites to gather data for the production of experimental statistics.
- Retrieval and assessment of information from financial statements filed with the National Court Register.

WP3 has established a sizeable base for systematic enterprise web-data acquisition

~900,000

Raw company records

Located in Poland.

>100,000

Accessible websites

Records returning HTTP 200

Multi-source

Content dimensions

Metadata + body content and
contact information

Acquisition chain

Company records



Website URL filter



HTTP verification



Content collection



Classification / use

Result so far: infrastructure and processes are being put in place for systematic acquisition, cleaning and indexing
- creating a foundation for activity classification and future statistical production.

WP3 and the new DAS and Datalab

Main issue: trainings provided in old datalab

Scraping in the new DAS (dozens of GBs)

Fixed with Eurostat Staff the issues related to the old OBEC scripts

Be aware of changes over time (positive changes – anonymization within DAS)

Local LLMs - non existing NACE codes (e.g. gemma), need RAG approach

Anonymize By text

Anonymize by URL

Simulate anonymization by Text

Language *

All language (... ▼)

Entities to Detect *

Credit Card
Crypto Wallet
Date Time
Email Address
IBAN Code
IP Address
Nationality/Religious/Political Group
Location
Person
Phone Number
Medical License
URL

LLM and NACE (PKD in Polish) codes

PROMPT (in Polish): Which PKD code would you assign to an organization with the following website content: "We sell a full range of construction supplies, from sand to garden architecture elements"?

PLLM (12B) – Polish LLM for Public Administration

Przykro mi ale nie jestem w stanie odpowiedzieć na twoje pytanie.

Może spróbujesz opisać je w inny sposób?

Bielik – Polish LLM for general purposes

Kodem PKD najbardziej pasującym do opisanej działalności jest ****47.59.Z**** - Sprzedaż detaliczna mebli, sprzętu oświetleniowego i pozostałych artykułów użytku domowego prowadzona w wyspecjalizowanych sklepach.

Jednakże, biorąc pod uwagę szczegóły dotyczące asortymentu budowlanego i elementów architektury ogrodowej, można również rozważyć kod ****47.59.Z**** w kontekście sprzedaży artykułów użytku domowego związanych z wyposażeniem ogrodu i budownictwem.

Jeśli firma specjalizuje się głównie w materiałach budowlanych, bardziej odpowiedni mógłby być kod ****46.73.Z**** - Sprzedaż hurtowa drewna i materiałów budowlanych oraz wyposażenia sanitarnego.

Ostateczny wybór zależy od tego, czy firma skupia się na sprzedaży detalicznej szerokiego asortymentu (w tym przypadku ****47.59.Z****), czy też specjalizuje się w materiałach budowlanych (wtedy ****46.73.Z****).

Warto również sprawdzić, czy nie ma bardziej szczegółowych kodów PKD, które lepiej odpowiadają specyfice działalności firmy.

Gemma

Do organizacji sprzedającej asortyment budowlany, obejmujący pełny zakres od piasku po elementy architektury ogrodowej, przypisałbym następujące kody PKD:

- * ****47.7.10** - Handel wyrobami budowlanymi** – To jest główny kod, który najlepiej opisuje działalność organizacji.
- * ****47.9.10** - Handel gruzem i żwirem** – Ponieważ sprzedaż piasku jest istotną częścią oferty.
- * ****47.7.20** - Handel materiałami izolacyjnymi** – Jeśli organizacja oferuje również materiały izolacyjne, które często są kupowane wraz z innymi materiałami budowlanymi.

****Dodatkowe uwagi:****

- * Warto sprawdzić, czy w ofercie organizacji znajdują się również elementy związane z architekturą ogrodową (np. gotowe rozwiązania, systemy nawadniania, itp.) i ewentualnie dodać kody PKD odnoszące się do tego segmentu.
- * Dokładny wybór kodów powinien być potwierdzony przez analizę konkretnych produktów i usług oferowanych przez organizację oraz w oparciu o wytyczne klasyfikatora PKD.

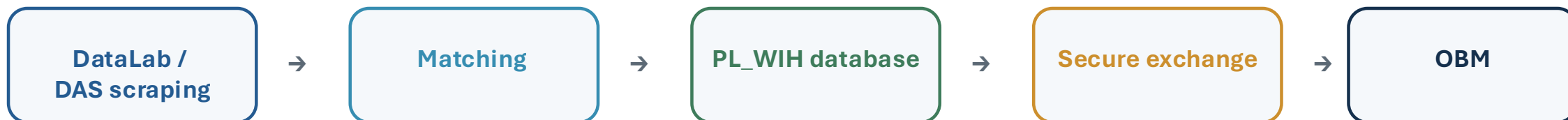
****Zastrzeżenie:**** Pamiętaj, że to tylko sugestie oparte na podanych informacjach. Ostateczny wybór kodów PKD powinien zostać dokonany po dokładnej analizie działalności organizacji.

WP4: Architecture with the secure data-flow foundation

Infrastructure already analysed or established



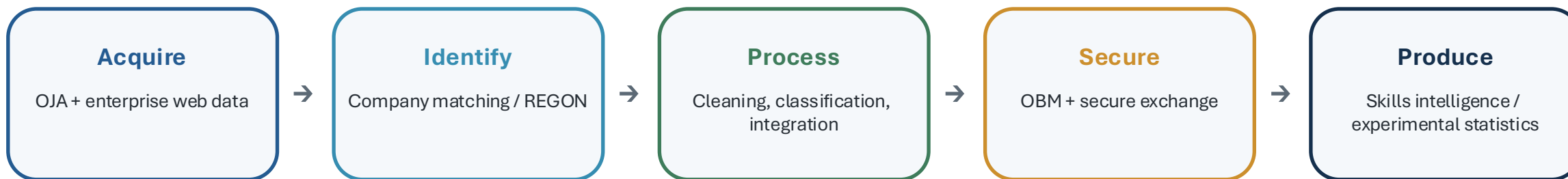
Current data-flow concept for WP2/WP3



WP4: Summary

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- 1) Analysis of existing infrastructure used by public statistics entities
 - 2) Analysis of already established systems for acquiring internet data
 - 3) Development of a data security plan for OJA/OBEC
 - 4) Creation of a roadmap for medium- and long-term implementation

The strongest achievement is the end-to-end implementation chain



Why this matters for implementation

Methodology

WP2 and WP3 demonstrates robust entity matching and a route to skills intelligence.

Infrastructure

WP4 are establishing scalable acquisition, processing and transfer capabilities.

Governance

WP1 keeps technical, organisational and financial activities aligned.

Implementation is progressing, with clear issues to resolve next

Data quality

Scraped structures and collected information require further cleaning and assessment.

Security & formal requirements

Data transfer to/from OBM must satisfy formal and security constraints.

Ambiguous entity matching

Different cases require manual verification.

Substantive linkage

The current OJA-ZUS integration is a technical feasibility demonstration; time-lag analysis is still needed.

Next implementation priorities are concrete and actionable

WP2

Analyse the time lag between OJA publication and appearance of the corresponding newly hired employee in ZUS data; advance substantive linkage of skills to occupations.

WP3

Continue website cleaning, NACE classification and automation of financial-statement collection.

WP4

Develop ETL packages, improve OBM transfer, organise code, introduce a dedicated scraping server and resolve security/data-quality issues.

WP1

Continue quarterly monitoring, cross-WP coordination and targeted capacity building to keep the implementation trajectory aligned.

Thank you!

Jacek Maślankowski, Statistics Poland
j.maslankowski@stat.gov.pl